

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Hackathon

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO5 –Hackathon	Assessment:	Assessment Tool 3– Hackathon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and interpret datasets using data preprocessing, statistical analysis, visualization, and machine learning techniques for effective data-driven insights and decision-making.		
Student Name:	P.Poorna Chandra	Reg. No.:	192472287
Section / Batch:	CSE – AI	Date:	29/08/26

Code of Conduct

I P.Poorna Chandra192472287 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P.Poorna Chandra

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	

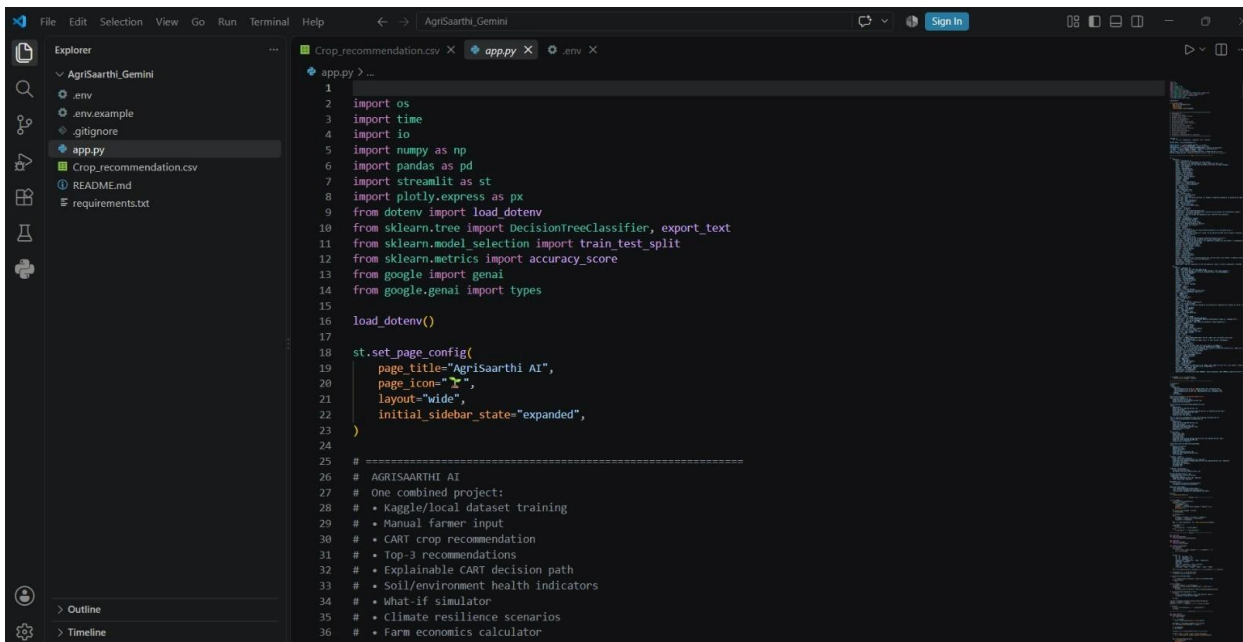
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	I n - d e p t h analysis with s t r o n g r e a s o n i n g and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandab le	Limited clarity and structure	Poor clarity; difficult to understand	

presentation

DSA0405 Fundamentals of Data Science

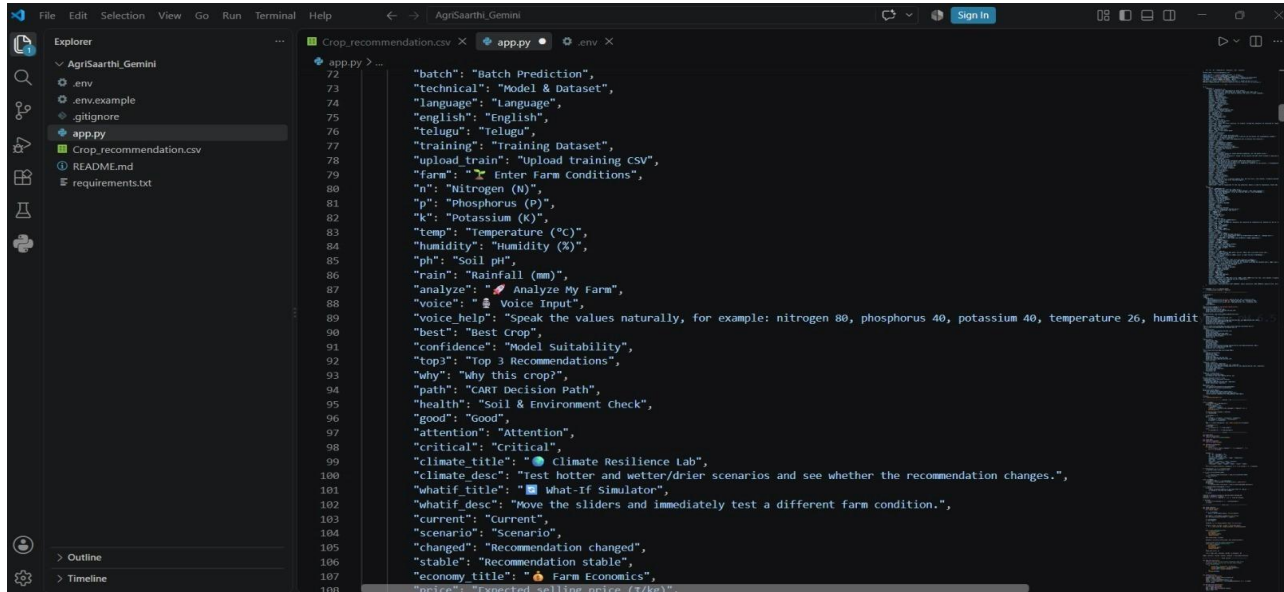
CO5-AT3 – Hackathon

Code:



The image shows a screenshot of a Visual Studio Code editor window. The Explorer panel on the left shows a project named 'AgriSaarthi_Gemini' with files: .env, .env.example, .gitignore, app.py, Crop_recommendation.csv, README.md, and requirements.txt. The main editor area displays the code for 'app.py'. The code imports various libraries including os, time, io, numpy, pandas, streamlit, plotly, dotenv, sklearn, and google.genai. It sets up a Streamlit page configuration and includes a list of project features.

```
1
2 import os
3 import time
4 import io
5 import numpy as np
6 import pandas as pd
7 import streamlit as st
8 import plotly.express as px
9 from dotenv import load_dotenv
10 from sklearn.tree import DecisionTreeClassifier, export_text
11 from sklearn.model_selection import train_test_split
12 from sklearn.metrics import accuracy_score
13 from google import genai
14 from google.genai import types
15
16 load_dotenv()
17
18 st.set_page_config(
19     page_title="AgriSaarthi AI",
20     page_icon="🌾",
21     layout="wide",
22     initial_sidebar_state="expanded",
23 )
24
25 # =====
26 # AGRISAARTHI AI
27 # One combined project:
28 # • Kaggle/local dataset training
29 # • Manual farmer input
30 # • CART crop recommendation
31 # • Top-3 recommendations
32 # • Explainable CART decision path
33 # • Soil/environment health indicators
34 # • What-if simulator
35 # • Climate resilience scenarios
36 # • Farm economics calculator
```



Implementation:

AgriSaarthi

Language / భాష

English ☒ Telugu ☐

పొలం డ్యాష్‌బోర్డ్

పొలం పివ్వాలి

పంట సిఫార్సు

ఏమైతే సమస్యలేటర్

వాతావరణ ప్రయోగశాల

పొలం ఆర్థిక లెక్కలు

AI సహాయకుడు

బ్యాచ్ అంచనా

మోడల్ & డేటాసెట్

API సిస్టం ఆన్

Training Dataset

Training CSV అప్‌లోడ్ చేయండి

ప్రతి రైతుకు AI ఆధారిత పంట నిర్ణయ సహాయం

అగ్రిసార్థి AI

మీ నేలను తెలుసుకోండి. వాతావరణాన్ని అర్థం చేసుకోండి. సరైన పంటను ఎంచుకోండి.
CART పంటను సూచిస్తుంది. నిజమైన AI ఫలితాన్ని సులభమైన భాషలో వివరిస్తుంది.

CART

Kaggle Dataset

Gemini

Climate Lab

Economics

English + తెలుగు

పొలం డ్యాష్‌బోర్డ్

73.9%
పరిశ్ల ఖచ్చితత్వం

2200
శిక్షణ రికార్డులు

22
పంట రకాలు

7
మోడల్ & డేటాసెట్

Complete farmer workflow

AgriSaarthi

Language / భాష

English ☒ Telugu ☐

Farm Dashboard

Farm Inputs

Recommendation

What-If Simulator

Climate Lab

Farm Economics

AI Assistant

Batch Prediction

Model & Dataset

API Ready

Training Dataset

Upload training CSV

AI-powered crop intelligence for every farmer

AgriSaarthi AI

Know your soil. Understand your climate. Choose the right crop.
CART predicts the crop. Real AI explains the result in simple language.

CART

Kaggle Dataset

Gemini

Climate Lab

Economics

English + తెలుగు

Enter Farm Conditions

Nitrogen (N)

30.00 - +

Temperature (°C)

26.00 - +

Rainfall (mm)

180.00 - +

Phosphorus (P)

37.00 - +

Humidity (%)

74.00 - +

Potassium (K)

36.00 - +

Soil pH

6.50 - +

Voice Input

Record

00:00

Speak the values naturally, for example: nitrogen 80, phosphorus 40, potassium 40, temperature 26, humidity 70, pH 6.5, rainfall 180.

 **AgriSaarthi**

Language / భాష

English ☒ Telugu ☐

☐ Farm Dashboard

☐ Farm Inputs

☒ Recommendation

☐ What-If Simulator

☐ Climate Lab

☐ Farm Economics

☐ AI Assistant

☐ Batch Prediction

☐ Model & Dataset

API Ready

Training Dataset

Upload training CSV

Deploy

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CART

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Climate Lab

Economics

English + తెలుగు

**Mungbean**
Best Crop

**100.0%**
Model Suitability

**Good**
Soil & Environment Check

Top 3 Recommendations

 **AgriSaarthi**

Language / భాష

English ☒ Telugu ☐

☐ Farm Dashboard

☐ Farm Inputs

☐ Recommendation

☐ What-If Simulator

☐ Climate Lab

☐ Farm Economics

☒ AI Assistant

☐ Batch Prediction

☐ Model & Dataset

API Ready

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Upload training CSV

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Climate Lab

Economics

English + తెలుగు

AgriSaarthi AI

Ask a question in English or Telugu. The AI explains the CART result instead of replacing it.

Ask your question

Why did you recommend this crop?
ఈ పంట ఎందుకు సిఫార్సు చేశారు?

Ask AgriSaarthi



Geotag:

